

PASHU SATHI — General & Solution Q&A

Smart India Hackathon 2026 | Internal Evaluation Viva Guide (General Domain)

Problem Statement: SIH26128 — Early Detection & Prevention of Livestock Diseases	Team: MOSAIC (6 Members)
Theme: MedTech / BioTech / HealthTech	Core Focus: Problem Overview, Solution Loop, Farmer UX & Impact

1. Core Problem & Rural Field Constraints

Q1: What exact gap in the current rural veterinary system does Pashu Sathi solve?

Role: Presenter / Pitcher

- **Physical Distance Barrier:** The nearest government veterinary dispensary is 15+ km away from smallholder farmers, creating reporting delays of 3 to 7 days.
- **Silent Epidemiological Spread:** Highly contagious infections like FMD and Lumpy Skin Disease spread between herds before paper reports reach district headquarters.
- **Catastrophic Household Loss:** A single cattle death wipes out up to 50% of a marginal farmer's yearly household income (approx. Rs. 75,000 net loss).

Q2: How will an illiterate rural farmer with no smartphone expertise use your app?

Role: Mobile App Dev / UI Specialist

- **Zero-Typing UI:** The interface eliminates text boxes; reporting is conducted through visual anatomical tap-icons (mouth, skin, udder, legs).
- **Multilingual Two-Way Voice AI:** Farmers simply press a prominent mic button and speak naturally in Marathi, Hindi, or English.
- **Passive Geo-Telemetry:** Exact GPS coordinates, date, and device timestamps are captured automatically in the background without user intervention.

Q3: Why won't farmers simply dial the existing 1962 veterinary toll-free helpline instead?

Role: Product & Systems Analyst

- **Bandwidth Limits:** 1962 call centers suffer from busy lines, limited call operators, and verbal descriptions without objective visual imagery.
- **No Spatial Intelligence:** Phone calls do not generate structured spatial-temporal outbreak coordinates on an interactive GIS map for state epidemiologists.
- **Complementary Backend:** Pashu Sathi can directly serve as the digital intake and ticketing software for 1962 call-center dispatchers.

2. Solution Workflow & Government Scheme Integration

Q4: Explain your 5-step rapid response loop in simple, non-technical terms.

Role: Healthcare Domain Specialist

- **Step 1 (Farmer Reports):** Farmer submits voice note or photo in under 30 seconds (works fully offline).
- **Step 2 (Preliminary Triage):** System flags suspected symptoms and alerts the assigned local block veterinarian.
- **Step 3 (Clinical Verification):** Licensed field veterinarian visits, scans animal QR passport, and verifies diagnosis.
- **Step 4 (Automated Spatial Clustering):** If 3 confirmed cases cluster within 25 km inside 48 hours, an outbreak cluster triggers.
- **Step 5 (Containment Deployment):** District animal husbandry officers receive GIS alerts and deploy targeted ring-vaccination.

Q5: How does Pashu Sathi align with government platforms like Bharat Pashudhan & INAPH?

Role: Regulatory & Policy Analyst

- **Intake Layer vs. Static Registry:** Bharat Pashudhan and INAPH are static census and animal tracking registries. Pashu Sathi is the real-time active surveillance engine.
- **Ear-Tag UID Compatibility:** Every animal record in Pashu Sathi directly links to the national 12-digit Pashu Aadhaar ear-tag standard.
- **Scheme Acceleration:** Directly accelerates DAHD's National Animal Disease Control Programme (NADCP) and LHDCP targets to eradicate FMD and Brucellosis by 2030.

3. Economic Impact & 30-Second Elevator Defense

Q6: What is the tangible ROI or financial justification for state animal husbandry departments?

Role: Impact & Economic Analyst

- **Rs. 20,000 Crore Loss Mitigation:** FMD alone causes over Rs. 20,000 Crore in direct losses annually; early containment saves milk yield and draught power.
- **Targeted Ring Vaccination:** Instead of expensive, delayed statewide blanket vaccination, officers direct cold-chain logistics strictly to the 25–50 km buffer zone.
- **Public Health Safety:** Fast isolation of Rabies, Anthrax, and Brucellosis mitigates dangerous zoonotic transmission to rural human populations.

30-Second Closing Elevator Pitch for Judges:

"Pashu Sathi transforms rural livestock healthcare from slow, reactive paper reporting into instant, digital syndromic surveillance. By empowering low-literacy farmers with offline voice reporting, enforcing licensed veterinary verification, and providing state officers with real-time GIS containment maps, we stop epidemics at 3 cases instead of hundreds—protecting smallholder livelihoods and the nation's agricultural economy."